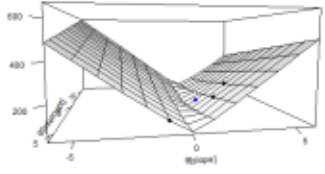
Introduction to Machine Learning

ML-Basics
Losses & Risk Minimization



Learning goals

- Know the concept of loss
- Understand the relationship between loss and risk
- Understand the relationship between risk minimization and finding the best model

Learning goals

- Know concept of loss function
- Understand concept of theoretical and empirical risk
- Understand relationship between risk minimization and finding best model

HOW TO EVALUATE MODELS

- Training a learner = optimize over hypothesis space
- Find function that matches training data best
- We compare point-wise predicted outputs to observed labels

When training a learner we optimize over the hypothesis space, to find the function which matches our training data best.

This means we are looking for a function, where the predicted output per training point is as close as possible to the observed label.

Features x		Target y
People in Office (Feature 1) x_1	Salary (Feature 2) x_2	Worked Minutes Week (Target Variable)
4	4300 €	2220
12	2700 €	1800
5	3100 €	1920

$\mathcal{D}_{\text{train}}$

$\approx \hat{\mathcal{R}}$

Prediction $\hat{y}$
Worked Minutes Week (Target Variable)
2588
1644
1870

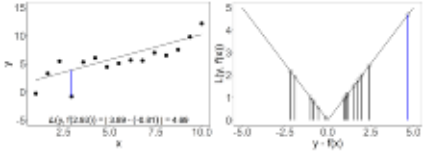
- To make this precise, we need to define now how we measure the difference between a prediction and a ground truth label pointwise.

LOSS

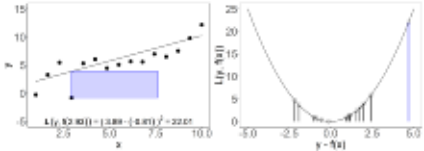
Loss function $L(y, f(\mathbf{x}))$ quantifies point-wise how we measure errors in predictions for a single $\mathbf{x}$:

$$L : \mathcal{Y} \times \mathbb{R}^g \to \mathbb{R}.$$

Regression: Could use absolute L1 loss $L(y, f(\mathbf{x})) = |y - f(\mathbf{x})|$:



or L2-loss $L(y, f(\mathbf{x})) = (y - f(\mathbf{x}))^2$:



RISK OF A MODEL

- Theoretical **risk** of a candidate model f is the **expected loss**
  (Theoretical) risk of a candidate model f is the expected loss $f(\mathbf{x})$
  measured by a loss function $L(y, f(\mathbf{x}))$ is the **expected loss**

$$\mathcal{R}(f) := \mathbb{E}_{xy}[L(y, f(\mathbf{x}))] = \int L(y, f(\mathbf{x}))\, \mathrm{d}\mathbb{P}_{xy}$$

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- Average error we incur when we use f on data from $\mathbb{P}_{xy}$
  This is the average error we incur when we use f on data from $\mathbb{P}_{xy}$.
- Goal in ML: Find a hypothesis $f \in \mathcal{H}$ that **minimizes** this
  Goal in ML: Find a hypothesis $f(\mathbf{x}) \in \mathcal{H}$ that **minimizes** risk.

Problems:

- $\mathbb{P}_{xy}$ is unknown
- Could estimate $\mathbb{P}_{xy}$ non-parametrically, e.g., by kernel density estimation, doesn't scale to higher dimensions
- Could efficiently estimate $\mathbb{P}_{xy}$, if we place assumptions on its form, e.g. cf. discriminant analysis

EMPIRICAL RISK

Problem: Minimizing $\mathcal{R}(f)$ over f is not feasible.

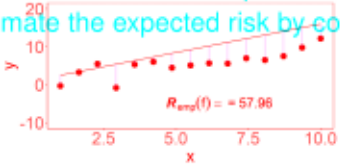
- Have n i.i.d. data from $\mathbb{P}_{xy}$, approximate expected risk empirically
- Just sum up all losses over training data

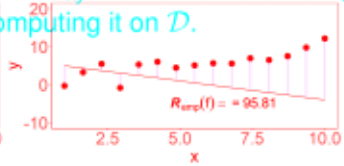
$$\mathcal{R}_{\text{emp}}(f) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

- Associates one quality score with each $f \in \mathcal{H}$
- Encodes: How well does f fits training data
- Now we get very close to solve this by optimization

$$\mathcal{R}_{\text{emp}} : \mathcal{H} \to \mathbb{R}$$

But as we have n i.i.d. data points from $\mathbb{P}_{xy}$ available we can simply approximate the expected risk by computing it on $\mathcal{D}$.





EMPIRICAL RISK MINIMIZATION

To evaluate, how well a given function f matches our training data, we now simply sum-up all f's pointwise losses.

- Best model = smallest risk
- For finite $\mathcal{H}$: we could tabulate exhaustively

$$\mathcal{R}_{\text{emp}}(f) = \sum^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right)$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	1	95.81
f_4	1	1.5	57.96

$$\mathcal{R}_{\text{emp}} : \mathcal{H} \to \mathbb{R}$$

This gives rise to the **empirical risk function** which allows us to associate one quality score with each of our models, which encodes how well our model fits our training data.

EMPIRICAL RISK MINIMIZATION

- The risk can also be defined as an average loss
- But usually $\mathcal{H}$ is infinitely large
- Instead: Simply consider risk surface w.r.t. the parameters $\boldsymbol{\theta}$

$$\bar{\mathcal{R}}_{\text{emp}}(f) = \frac{1}{n}\sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)}\right)\right).$$

The factor $\frac{1}{n}$ does not make a difference in optimization, so we will consider $\mathcal{R}_{\text{emp}} : \mathbb{R}^d \to \mathbb{R}$ most of the time.

- Since f is usually defined by **parameters** $\boldsymbol{\theta}$, this becomes:

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
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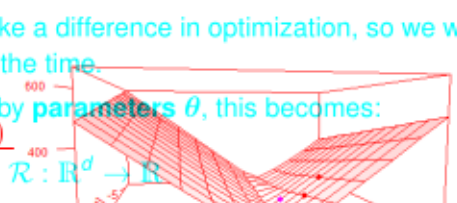

$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \sum_{i=1}^{n} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

EMPIRICAL RISK MINIMIZATION

The best model is the model with the smallest risk.

If we have a finite number of models f, we could simply tabulate them and select the best.

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
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EMPIRICAL RISK MINIMIZATION

But usually $\mathcal{H}$ is infinitely large.

Instead we can consider the risk surface w.r.t. the parameters θ.
(By this I simply mean the visualization of $\mathcal{R}_{\text{emp}}(\theta)$)

$$\mathcal{R}_{\text{emp}}(\theta) : \mathbb{R}^d \to \mathbb{R}.$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
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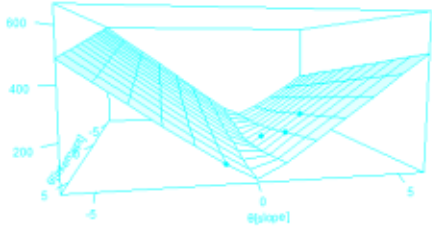


EMPIRICAL RISK MINIMIZATION

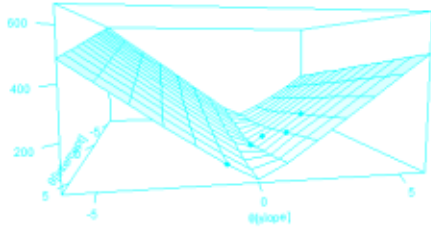
Minimizing this surface is called **empirical risk minimization** (ERM).

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \mathcal{R}_{\text{emp}}(\theta).$$

Usually we do this by numerical optimization.

$$\mathcal{R} : \mathbb{R}^d \to \mathbb{R}.$$

Model	$\theta_{intercept}$	θ_{slope}	$\mathcal{R}_{\text{emp}}(\theta)$
f_1	2	3	194.62
f_2	3	2	127.12
f_3	6	-1	95.81
f_4	1	1.5	57.96
f_5	1.25	0.90	23.40



In a certain sense, we have now reduced the problem of learning to **numerical parameter optimization**.